

Systematic FX trading with regression learning and transaction cost analysis

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Jupyter Notebook

Regression-based statistical learning is a convenient and transparent method for combining trading factors into composite signals. Sequential statistical learning considers only the data available at each time point to choose and parameterize the "best" model and to generate signals without hindsight bias. Yet assessing PnL potential in backtests also requires estimates of transaction costs as a function of the notional amounts traded. Such estimates can be generated by (i) using LLM prompts to retrieve key cost parameters and (ii) feeding them into appropriate Python functions for cost imputation.

This article illustrates regression-based learning and transaction cost analysis for systematic developed-market FX forward trading with macro factors. The sequential learning process generates signals with highly significant predictive power. Risk-adjusted returns may look modest, but they have been uncorrelated with major market benchmarks, and the statistical probability of long-term value generation is very high. Moreover, transaction cost analysis attests to the scalability of a low-frequency strategy in liquid markets, showing very large average annual PnLs over time.

The article below is based on Macrosynergy's proprietary research. Please quote as "Sueppel, Ralph, Swandel, Stefan, and Tyagi Palash, 'Systematic FX trading with regression learning and transaction cost analysis', Macrosynergy research, June 2026."

This is a materially revised and extended version of the post "[FX trading signals with regression-based learning](#)" of April 23, 2024.

The attached Jupyter notebook allows for the audit and replication of the research results. The notebook's operation requires access to J.P. Morgan DataQuery to download data from JPMaQS. Access is generally available to DataQuery clients, subject to some restrictions. J.P. Morgan offers unrestricted access for institutional clients, including free trials. Also, an [academic support program](#) sponsors data sets for research projects.

The case for sequential regression-based learning

Regression-based sequential statistical learning provides a practical framework for combining candidate factors into a unified trading strategy signal. The approach selects models and hyperparameters sequentially, using only information available at the time and avoiding any look-ahead bias. On each trading day, the current "best" model is used to generate signals for the following period. These signals, therefore, provide a sound basis for realistic backtesting, and the same

procedure can be applied directly to future live trading. The fundamentals of regression-based learning were introduced in **Macro trading signals with regression-based machine learning**. The effectiveness of regression-based learning owes to convenience and simplicity.

- **Convenience** means that regression models automatically assign weights to candidate trading factors based on past statistical relationships. Many regression model types, such as non-negative least squares and elastic net, can also exclude factors from signal formation when they exhibit insufficient or implausible predictive power.
- **Simplicity** means that standard linear regression models are easy to understand and interpret. A linear regression coefficient measures the expected change in the target return associated with a one-unit increase in a predictor variable, holding all other predictors constant. If the predictor variables are normalized, the coefficient measures the expected change in the target return resulting from a one-standard-deviation change in the predictor.

Regression-based learning is particularly useful for handling a broad set of conceptually distinct and potentially complementary factors. Theory can help identify such factors, but it is often silent on how they should be weighted. Factor weighting is especially important for systematic trading strategies in foreign exchange, where returns are influenced by a wide range of economic trends.

To select and weight factors, it is often useful to apply **panel regression**, a statistical model that estimates relationships between returns and factors by considering contracts across multiple currency areas and time periods simultaneously. Panel regression broadens the scope and diversity of historical information by accounting for both intertemporal and cross-country variation in factors and target returns. If parameters are constrained to be equal across currency areas, it can also serve as a form of regularization, i.e., a technique that constrains a model's flexibility to reduce overfitting,

Importantly, regression models can be enhanced by specialized modifications geared towards generating better trading signals:

- **Reliability adjustment** scales regression coefficients by their statistical precision. Simply put, before calculating the trading signal, one divides the macro factor coefficients by their estimation standard errors. As data panels become larger, parameter estimation errors decline. The adjustment aligns intertemporal risk-taking with signal reliability ("**How to adjust regression-based trading signals for reliability**").
- The **"global-local" method** estimates coefficients in an international data set by considering both global and country-specific relations. It is a gentle form of regularization, shrinking country-specific ("local") coefficients towards their pooled ("global") values, thereby balancing the rigidity of the strict panel approach and the volatility of single-country regressions ("**Machine learning for international trading strategies**").
- **Dimension reduction methods**, such as principal component analysis and partial least squares, can improve regression-based learning by mitigating the effects of highly correlated regressors, which can make coefficients unstable, and by reducing overfitting through a focus on the most important and reliable sources of variation ("**Macro trading factors: dimension reduction and statistical learning**").

A simple approach to transaction cost imputation

Transaction costs here refer to the bid-offer spreads paid when entering and exiting positions ("trading costs") and when extending contract maturities ("roll costs"). Accounting for these costs improves strategy evaluation by providing a more realistic estimate of actual economic value in a specific institutional setting and by indicating the scalability of an idea. It also reveals the cost of strategy modifications, such as volatility targeting.

Transaction costs vary over time, depending on structural market developments and prevailing market conditions, such as liquidity, price volatility, and risk aversion. They also depend on execution methods. Here, however, we quantify their effects on a regression-based FX forward strategy in the simplest plausible way: by applying static, prompt-generated cost parameters to position changes and roll dates. The advantage of this approach is that it is easy to interpret, replicate, and modify.

We queried estimates of one-way execution costs for trading and rolling 1-month FX forwards for developed-market currency pairs at present via a ChatGPT 5.5 prompt (see **Annex 1**). In particular, we collect information on typical (median) and large (90th percentile) trading ticket sizes. Then we collect estimates of one-way execution costs for trading and rolling (about half of the bid-offer spreads) for these sizes. The information serves as the basis for two transaction cost functions for each currency pair: one for trading and one for rolling.

One-way trade execution and roll costs							
Currency	Trading pair / basket	Median size	Exec. cost at median	Roll cost at median	90th pct. size	Exec. cost at 90th pct.	Roll cost at 90th pct.
JPY	USD/JPY	\$50m	0.8 bps	0.3 bps	\$250m	1.3 bps	0.6 bps
CAD	USD/CAD	\$40m	1.0 bps	0.4 bps	\$200m	1.6 bps	0.7 bps
GBP	50% EUR/GBP, 50% GBP/USD	\$40m	1.0 bps	0.4 bps	\$200m	1.7 bps	0.8 bps
AUD	AUD/USD	\$35m	1.2 bps	0.5 bps	\$175m	2.0 bps	0.9 bps
CHF	EUR/CHF	\$30m	1.3 bps	0.5 bps	\$150m	2.2 bps	1.0 bps
SEK	EUR/SEK	\$20m	2.0 bps	0.8 bps	\$100m	3.3 bps	1.5 bps
NOK	EUR/NOK	\$15m	2.5 bps	1.0 bps	\$75m	4.2 bps	2.0 bps
NZD	NZD/USD	\$15m	2.3 bps	0.9 bps	\$75m	3.8 bps	1.8 bps

The ticket size and cost information must be mapped to position changes and position rolls on specific dates, both of which reflect the trading strategy and the assets under management (AUM). The **Macrosynergy package** handles this transaction-cost imputation and analysis via its **ProxyPnL** class. It translates raw trading signals into notional positions and then computes dollar-denominated profit and loss. There are various ways to turn normalized signals into position sizes. In this article, we simply use the ``dollar_per_signal`` parameter, which specifies a USD position size per contract signal. Other options include setting leverage relative to assets under management or targeting volatility in USD terms. However, these are more invasive procedures, as they change the nature of the strategy and should therefore be discussed separately.

Furthermore, the class offers two options for imputing trading and roll costs: static and time-varying. In both cases, costs are estimated based on size and percent cost estimates for median and large trade tickets, which serve as the parameters of a rectified linear function. For trading costs, the function is for each currency pair as follows

$$C_{\Delta P} = \left(c_{med} + \max(0, \Delta P - \Delta P_{med}) \cdot \frac{c_{90pct} - c_{med}}{\Delta P_{90pct} - \Delta P_{med}} \right) \cdot \Delta P$$

where

- ΔP is the position change or ticket size for which trading costs are calculated,
- ΔP_{med} and ΔP_{90pct} are the median and 90th percentile ticket sizes for the currency pair,

- $C_{\Delta P}$ is the trading cost in USD for the position change, and
- c_{med} and c_{90pct} are the median and 90th percentile size-dependent trading costs of the currency pair in percent of the ticket size, reflecting half of the bid-offer spread in the market.

This means that small position changes, up to the median trade size, are assigned the median trading cost per dollar. For larger changes, costs rise in proportion to trade size, based on how much costs increase between the median and the 90th percentile trade size. Costs rise faster when liquidity limits are reached more easily.

The cost function for rolls is similar:

$$CR_P = \left(cr_{med} + \max(0, P - \Delta P_{med}) \cdot \frac{cr_{90pct} - cr_{med}}{\Delta P_{90pct} - \Delta P_{med}} \right) \cdot P$$

where

- P is the “untraded” position on a roll date, i.e., a position that was carried over from the previous trading date and that is not liquidated on that date, thus requiring maturity extension,
- CR_P is the roll cost in USD for the maturity extension, and
- cr_{med} and cr_{90pct} are the median and 90th percentile size-dependent roll costs of the currency pair in percent of the ticket size. They are based on typical trading sizes, not roll sizes, as the latter data are harder to procure.

Calculations for time-varying transaction costs simply use time series of the median and the 90th percentile size and per-dollar cost estimates. However, generating the corresponding time series would exceed the scope of this article.

FX markets and plausible trading factors

We consider macroeconomic trading factors for directional FX forward positions across eight developed countries relative to their natural benchmarks. They encompass all liquid and floating developed markets’ currencies, excluding the U.S. and the euro area, i.e.:

- the Australian dollar (AUD), the Canadian dollar (CAD), the Japanese yen (JPY), and the New Zealand dollar (NZD), all versus the U.S. dollar,
- the Swiss franc (CHF), the Norwegian krone (NOK), and the Swedish krona (SEK), all versus the euro, and
- the British pound (GBP) versus an equally-weighted basket of U.S. dollar and euro.

More specifically, we focus on volatility-targeted 1-month FX forward positions that are rolled and rebalanced monthly. For more details on the generic return calculation, view documentation [here](#).

Macroeconomic theory suggests a wide range of indicators that may help predict the performance of developed-market currencies. These indicators mainly reflect economic competitiveness, a country’s attractiveness to capital flows, and the outlook for monetary policy. Their predictive power can be understood through the theory of rational inattention, which

argues that market participants cannot continuously process all available information. Instead, they rationally focus on a limited set of key indicators ("**Rational inattention and trading strategies**"), leaving scope for value generation by a cost-efficient, systematically informed trading system.

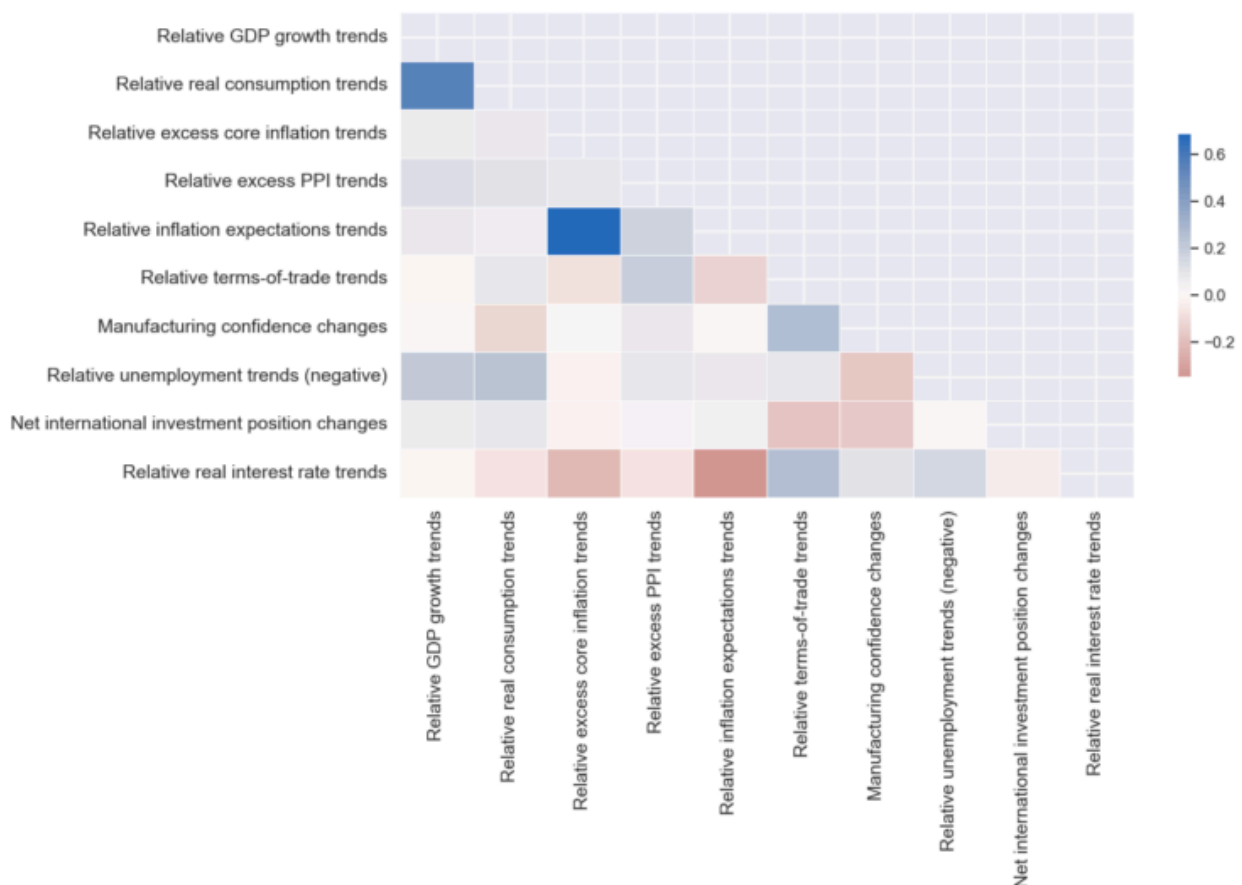
To measure predictive power, we need point-in-time economic data that can represent the market's information state. These are available from the *J.P. Morgan Macrosynergy Quantamental System* (JPMaQS). An information state is the latest instance of an economic indicator based on the data vintage available on a given day. It is also called a macro-quantamental indicator. Quantamental indicators can be aggregated to yield point-in-time macro-quantamental factors that capture broad, relevant economic developments. Here, we focus on a small set of factors that are particularly popular in FX trading. They are explained in detailed in **Annex 2** and can be summarized as follows:

- **Relative GDP growth trends:** This is the difference between nowcast GDP growth in the local currency area and the benchmark currency area. Higher relative growth indicates better economic competitiveness and warrants tighter monetary policy.
- **Relative private consumption growth:** This measures the strength in real household spending in the base economy versus the benchmark currency area. Stronger real spending argues for tighter monetary policy.
- **Manufacturing confidence improvement:** These are changes in normalized headline measures of local-currency area manufacturing business sentiment. Generally, improving confidence in tradable goods industries signals stronger competitiveness and leads a positive news flow.
- **Relative unemployment trends:** These are recent changes in seasonally adjusted unemployment rates in the local economy versus the benchmark currency area. Relative tightening of the labor market in the local currency area bodes for relative tightening of monetary policy.
- **Relative excess core CPI inflation:** This factor measures the excess of local core inflation versus the excess of core inflation in the benchmark currency area. Excess here means relative to the central bank's inflation target. If local inflation exceeds its target more than in the benchmark currency area, it is more likely that the local central bank will run a tighter monetary policy and tolerate currency strength.
- **Relative excess producer price inflation:** This is the difference between the excess producer price trend in the local economy and the benchmark currency area. Higher local currency area excess producer price growth indicates both pressures for tighter monetary policy and improving competitiveness.
- **Relative excess inflation expectations:** This is the difference between excess estimated inflation expectations in the local economy and the benchmark currency area. Higher relative inflation expectations across similar economies typically translate into tighter monetary policy and greater tolerance for currency strength.
- **Relative real interest rates:** This is the difference between various real interest rate in the local market and in the benchmark currency area. Higher real interest rates are indicative of higher implied central bank subsidies (or lower penalties) and higher risk premia.
- **International investment position trends:** Conceptually, these are changes in recorded external financial relations between residents and non-residents over the medium term. A rapid increase in net liabilities indicates large past capital inflows that may be hard to sustain and, hence, increases the risk of subsequent currency weakness.
- **Terms-of-trade dynamics:** These are changes in export prices relative to changes in import prices. Improving terms of trade often precede economic outperformance, capital inflows, and positive economic news, all of which support currency strength.

The above categories have been chosen and calculated based on theoretical priors, not statistical optimization. For convenience, all constituent candidates are given the “right sign” that makes their theoretically expected predictive direction positive. Moreover, all signal constituent candidates are sequentially normalized around their zero values, i.e., each indicator is divided by its standard deviation across past panels, thereby avoiding look-ahead bias. The resultant “factor scores” are also winsorized (capped or floored) at three standard deviations.

Correlation across factors within the panel of eight developed markets has mostly been modest or even negative since 2000. However, there have been exceptions: relative excess core inflation versus relative inflation expectations, and relative real GDP growth versus relative real consumption recorded more than 50% correlation. This is a reminder of a simple truth: conceptually different factors can be highly correlated. Multivariate linear regression addresses multicollinearity by estimating coefficients based on each predictor's independent explanatory power. Correlation does not bias coefficient estimates. However, if collinearity becomes very high, linear regression struggles to separate the effects of the correlated variables, leading to larger standard errors and unstable coefficients.

Monthly cross correlation of signal constituent candidates for a panel of 8 DM countries since 2000



A simple regression-based learning approach

To combine the 10 factors into a single signal without hindsight, one can use a sequential regression-based statistical learning process. Starting with a minimum sample size (here, we require 3 years for at least two cross-sections), we determine an optimal regression model at the end of each month from a grid of variations, using cross-validation. Then we use the full concurrent history to estimate the model's parameters and predict next month's return. This prediction is our raw signal.

The **SignalOptimizer** class is the central engine in the Macrosynergy learning module for building and maintaining optimized trading signals from panel quantamental data sequentially. It takes a panel of lagged macro-financial features and target returns, repeatedly trains and evaluates a user-specified set of machine-learning models and hyperparameter configurations using panel-aware cross-validation. It then selects the best-performing model according to chosen scoring metrics and generates out-of-sample forecasts. The process expands the training sample as new observations become available. Beyond producing forecasts with the **calculate_predictions** method, it stores the history of model selections, hyperparameters, coefficients, intercepts, and feature-selection outcomes, enabling analysis of how the forecasting process evolves over time.

For a simple, systematic macro FX strategy, this approach requires the following preparatory steps, which are demonstrated in the accompanying Jupyter notebook:

- **Transform features and targets into suitable formats for scikit-learn:** It is important to remember that the features and targets of the analysis are panel data, i.e., two-dimensional datasets that contain one category across time and relevant countries of currency areas. These data are transformed into a (pandas) data frame for all features (X) and a double-indexed series of targets (y). The index dimensions are cross-section (currency area) and time. Frequency is reduced from daily to monthly by using the month-end value of the features and the cumulative value of the target returns. Finally, the features are lagged by one period.
- **Define appropriate cross-validation splitters:** Cross-validation methods for panels have been summarized in **Macro trading signal optimization: basic statistical learning methods**. Here, we use two training data splitter classes of the Macrosynergy package: The first is the **RollingKFoldPanelSplit** class, which instantiates splitters of temporally adjacent panel training sets that can border the test set from the past and future.



The second is the **ExpandingKFoldPanelSplit** class, which creates a panel analogue of scikit-learn's **TimeSeriesSplit** class where blocks of panel training sets always precede test sets chronologically and where the time span of the training sets increases.



For both methods, the number of splits increases with sample length.

- **Define appropriate scores for cross-validation:** To measure the success of models to predict unseen FX returns, we use a combination of two scorers. The **sortino_ratio** scorer produces a ratio of downside risk-adjusted returns of a simple strategy where the trader goes long when the predictions are positive by a single unit and short by a single unit when the predictions are negative. The **regression_balanced_accuracy** scorer calculates the average accuracy of positive and negative monthly predictions. The **calculate_predictions** method handles multiple scorers by first evaluating each scorer across CV test folds, then normalizing scores across model/hyperparameter candidates. Thereby, different scorer units become comparable. It then forms a composite score by averaging the adjusted and normalized scores across all scorers.
- **Definition of candidate models and hyperparameters:** The information is collected in three Python dictionaries of regression models and their hyperparameter grids. These can then be passed on to the appropriate scikit-learn classes and methods or their Macrosynergy package wrappers. The specific grids used are described below.

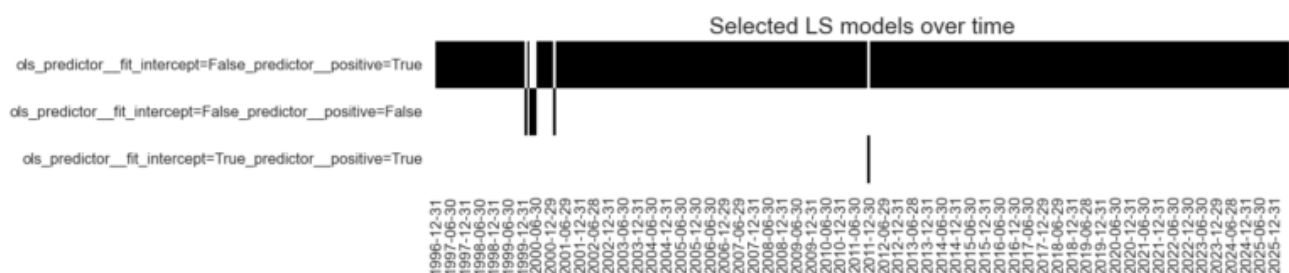
The sequential learning process implies that signals vary over time not only because economic indicators change, but also because model specifications and parameters are updated. The resulting sequence of optimized signals provides a valid basis for the out-of-sample evaluation of the overall learning process, using both statistical criteria and profit-and-loss (PnL) time series.

Empirical findings of three regression-based learning methods

For this article, we coded regression-based learning processes for three basic models separately, rather than jointly. This saves runtime and contains signal variation due to model changes, which can never help predictive quality. In this section, we summarize the regression methods, the (small) hyperparameter grids, and the characteristics of the optimal models over time.

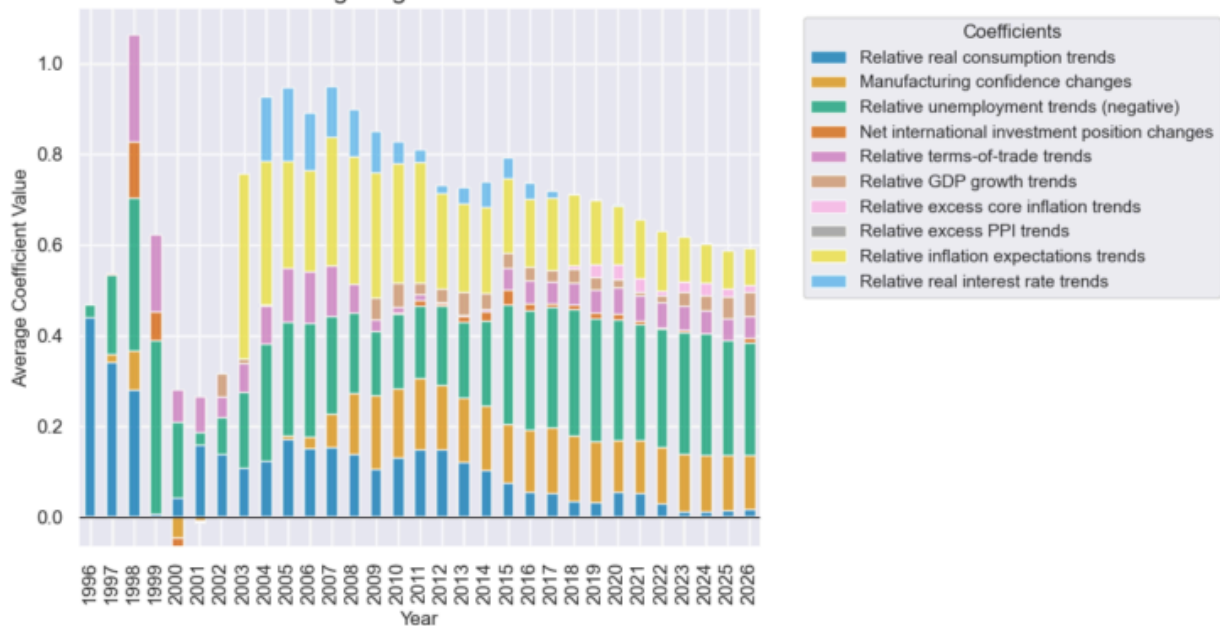
The **standard least-squares** panel models include ordinary least squares and non-negative least squares regressions. Non-negative least squares (NNLS) is a variant of ordinary least squares (OLS) that estimates regression coefficients by minimizing the sum of squared residuals subject to the constraint that all coefficients must be non-negative. It effectively eliminates all factors in the prediction process with negative coefficients, i.e., predictive directions that are inconsistent with the theoretical priors. The hyperparameter grid offers the choice between inclusion and exclusion of a regression intercept (which should be zero if theoretical priors are correct) and between free coefficient values and non-negativity constraints.

The learning process displayed a clear preference for exclusion of intercepts and coefficient non-negativity. This means it supported the regression version that relied most on theoretical priors.



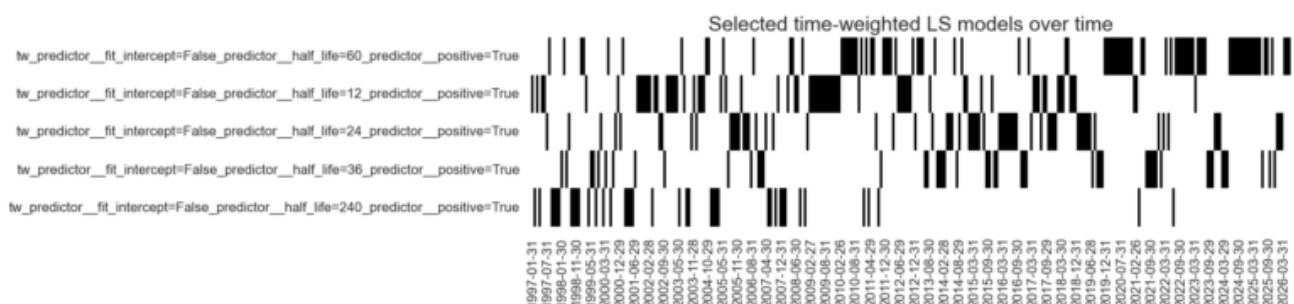
Factor coefficients changed materially during the early history of the signal generation process but stabilized with the expanding data sample and emphasized similar factors since the late 2000s. The factors with the highest weights have been relative unemployment trends, manufacturing confidence changes, relative inflation expectations trends, relative GDP growth trends, and relative terms-of-trade trends.

Optimal OLS/NNLS models' average regression coefficients for normalized features



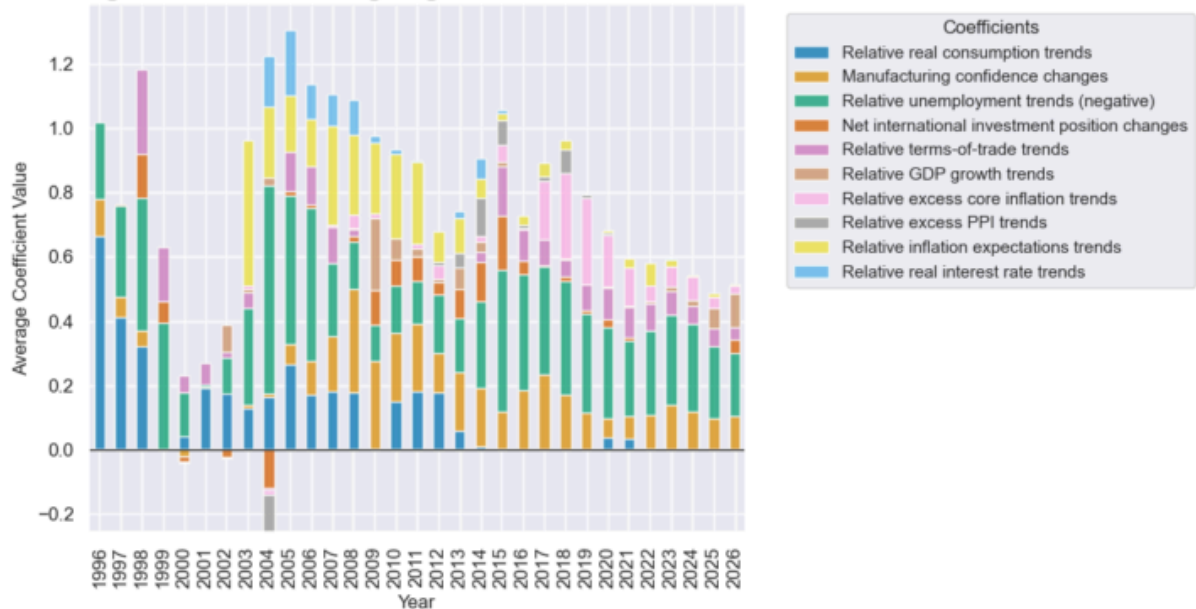
Time-weighted least squares (TWLS) is a regression method that assigns different weights to observation periods when estimating the regression coefficients, giving more weight to recent data and less weight to older data in accordance with exponential decay. The coefficients are chosen to minimize a weighted sum of squared residuals, so observations with larger weights have a greater influence on the fitted model. This approach assumes that relationships go through changes and seasons and that recent experiences are more valuable. Some of the hyperparameters of the learning process are the same ones that were used in standard least squares: intercept inclusion and non-negativity. However, we now also need to consider hyperparameters for the half-life of the time decay of observation weights, which we set between 1 year and 20 years.

As for standard least squares, the learning process preferred the specifications that accord with theoretical priors, setting intercepts to zero and enforcing positive coefficients. However, the process failed to settle on a specific decay ratio, mostly shifting between 2 years, 3 years, and 5 years of half-life.

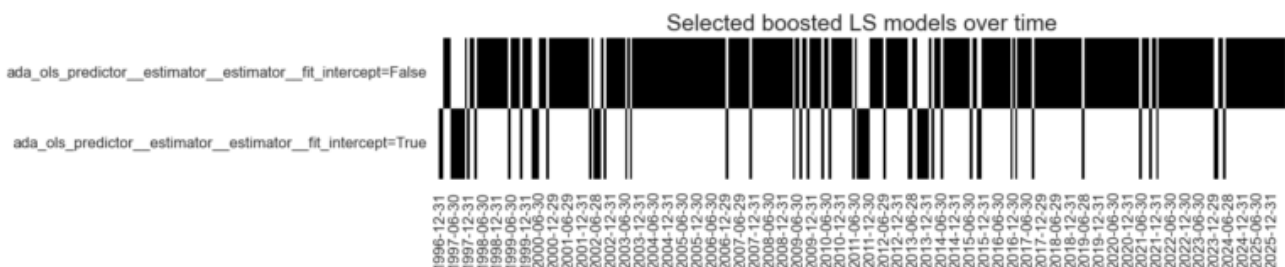


Due to time-weighting and model shifts, the time-weighted regression-based learning process did produce more changeable coefficient weights. The key factors in recent years were similar but not identical to the standard least squares model. Also, there has been a more pronounced decay of overall coefficient size in recent years, reflecting increasing preference for a longer (5-year) lookback half-life.

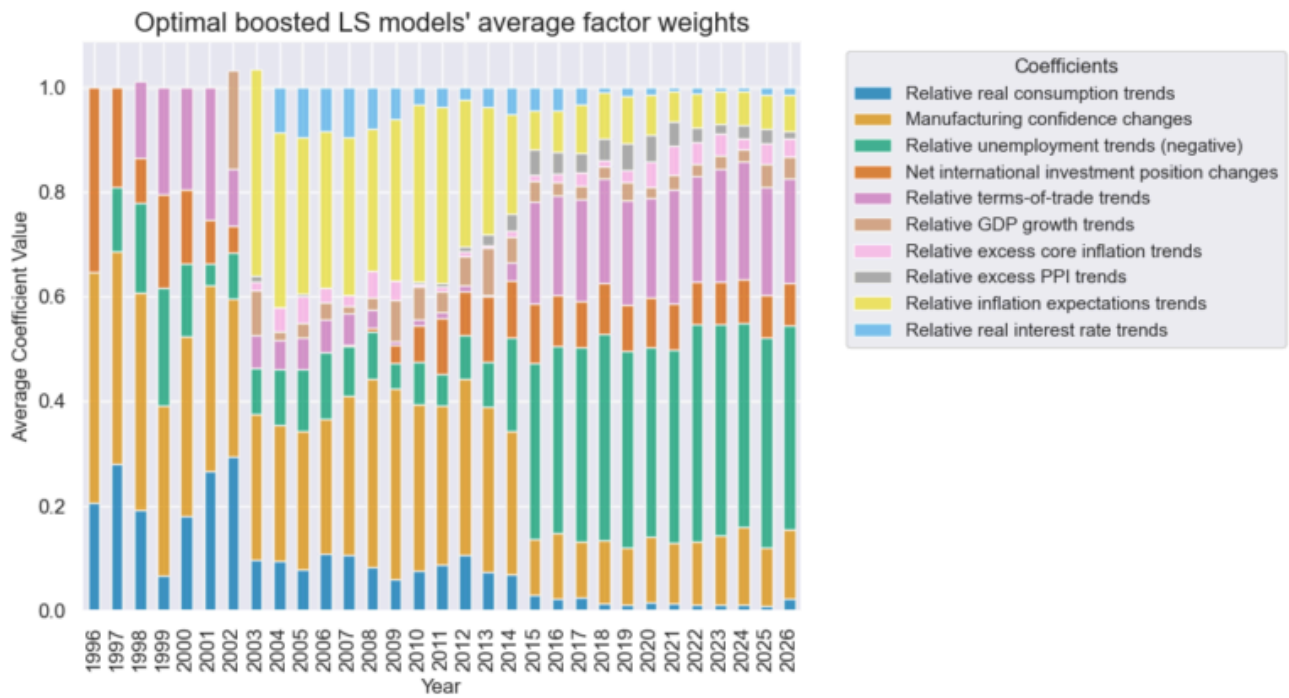
Optimal time-weighted LS models' average regression coefficients for normalized features



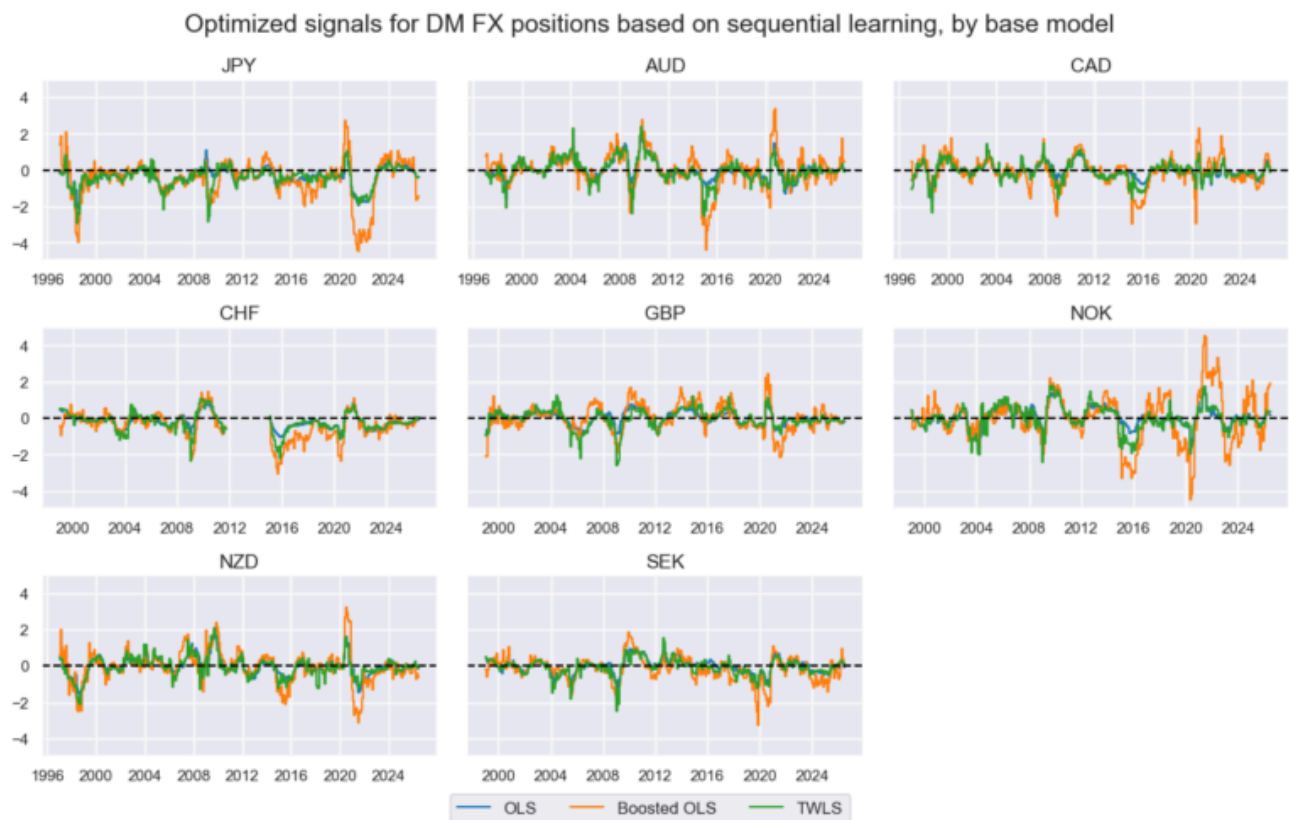
Panel regression with Adaboost combines OLS regression and ensemble learning. AdaBoost (Adaptive Boosting) is a standard method that combines multiple weak learners sequentially by adjusting the weights of data points after each model. The final prediction is a weighted combination of the individual learners, allowing the model to capture non-linear relationships and interactions that would be difficult for a single weak learner to identify. The application of panel regression with Adaboost for generating macro trading signals is explained in the article **Boosting macro trading signals**. The only hyperparameter is whether to include the intercept in the weak regression models. For the most part, the learning process prefers to forfeit intercepts.



The chart below visualizes the relative feature importances based on impurity measures. As the sample expanded, the most influential factors have become relative unemployment trends, relative terms-of-trade trends, changes in manufacturing confidence, and changes in the net international investment position.

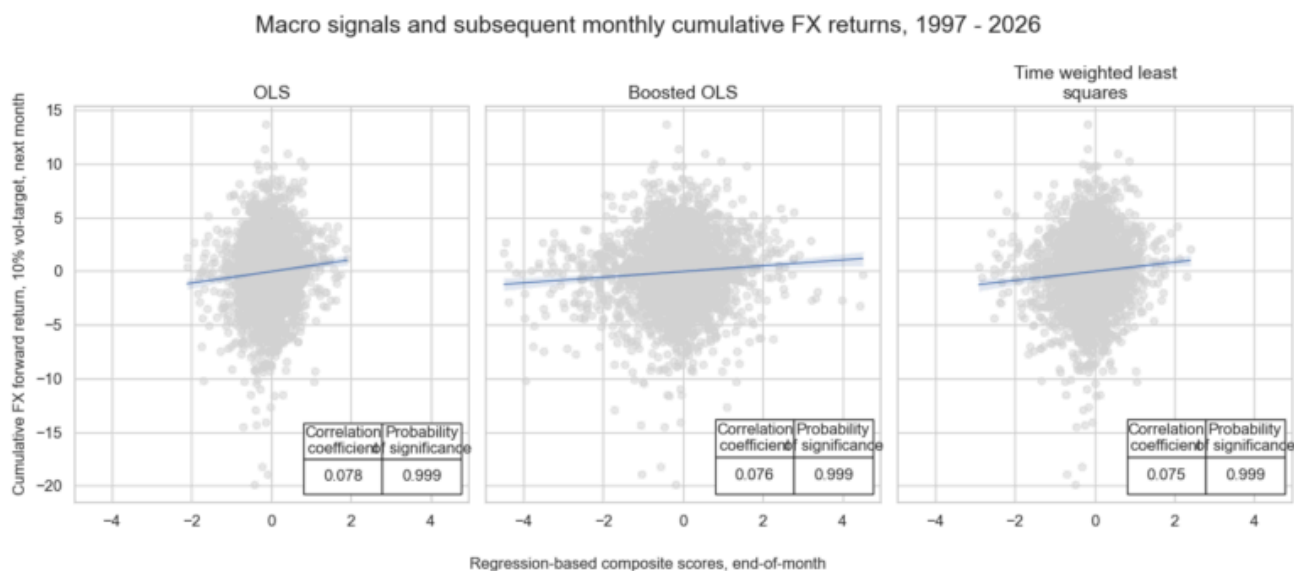


The sequentially optimized signals of three basic model types are plotted below. Standard and time-weighted least squares models have produced very similar signals. Boosted regression signals exhibit greater variation but follow similar cycles. All signals have fluctuated around the zero neutral line for all currencies in multi-year cycles with some additional short-term volatility.



Predictive power, simulated PnLs, and transaction costs

Panel regression analysis helps assess the strength and significance of the linear relation between month-end macro signals and next month's returns for the full panel of eight developed market FX forwards from 1997 to 2026 (May). It looks at both intertemporal and cross-currency variations in signals and returns. For all currencies, the monthly forward correlation has been positive, between 7 and 8%, and the probability of the relationship being systematic, not due to chance, has been 99.9% according to the **Macrosynergy panel test**. The significance of positive non-parametric forward correlations has been similarly high. Furthermore, the balanced accuracy of the monthly signal, i.e., the average accuracy of positive and negative return predictions, has been 51-52% for the three signal types.



We simulate the PnL evolution of simple trading strategies based on the optimal signals by using the **ProxyPnL** class of the Macrosynergy package. Unlike the **NaivePnL** class, which is used in most Macrosynergy research articles, here the approximated PnLs are defined on the notional positions of all underlying contracts and thereby allow consideration of transaction costs that depend on position and traded ticket sizes.

The trading strategy used here is simple. Each currency signal is normalized and capped at four standard deviations. At the beginning of each month, the strategy takes positions based on the signals from the end of the previous month, allowing one day for trading slippage. Positions are set at USD 50 million for each one-standard-deviation signal, either long or short the local currency against the benchmark currency. Total risk capital is USD 250 million. This means that leverage and value-at-risk change with signal strength, reflecting that opportunities for this simple systematic macro strategy vary over time.

Developed market FX forward notional positions for various regression-based signals, USD million



All regression-based signals have generated respectable risk-adjusted returns in the long run, given the simplicity and the small number of correlated contracts used:

- The **standard least-squares** signal has generated long-term (1997-2026) Sharpe and Sortino ratios of 0.6 and 0.8, respectively, before transaction costs. The Sharpe stability ratio has been 2.7, indicating an over 99% probability of generating systematic risk-adjusted returns (see [The Sharpe stability ratio of trading strategies](#)). The annualized return would have been 6.1% for an annualized standard deviation of 10.7%. This means the average annual PnL would have been USD 15 million. Correlations with the returns of key market benchmarks, such as the S&P500 future, the 10-year Treasury, and EURUSD forwards, have all been near zero. Transaction costs for that trading book would have reduced the annualized return to 5.8%, i.e., by about 0.3% or USD 750,000 per year.
- The numbers for the **time-weighted least-squares signal** would have been very similar. Sharpe and Sortino ratios reached 0.6 and 0.8, respectively. The average annual PnL would have been very close to USD15 million. The Sharpe stability ratio was slightly lower at 2.1 but still highly significant. All benchmark correlations were near zero. The transaction cost would have been slightly higher than that of the standard least-squares signal, at 0.34% or USD 850,000.
- The **boosted regression signal** would have produced both higher PnLs and larger transaction costs due to bolder position taking. Sharpe and Sortino ratios were 0.6 and 0.9, with the Sharpe stability ratio at 2.2. Annual average returns would have been 7.4% or USD18.5 million before transaction costs. Again, correlations with all major market benchmarks were near zero. Transaction costs were slightly above USD1.1 million per year, reducing annualized returns by 0.4-0.5%.

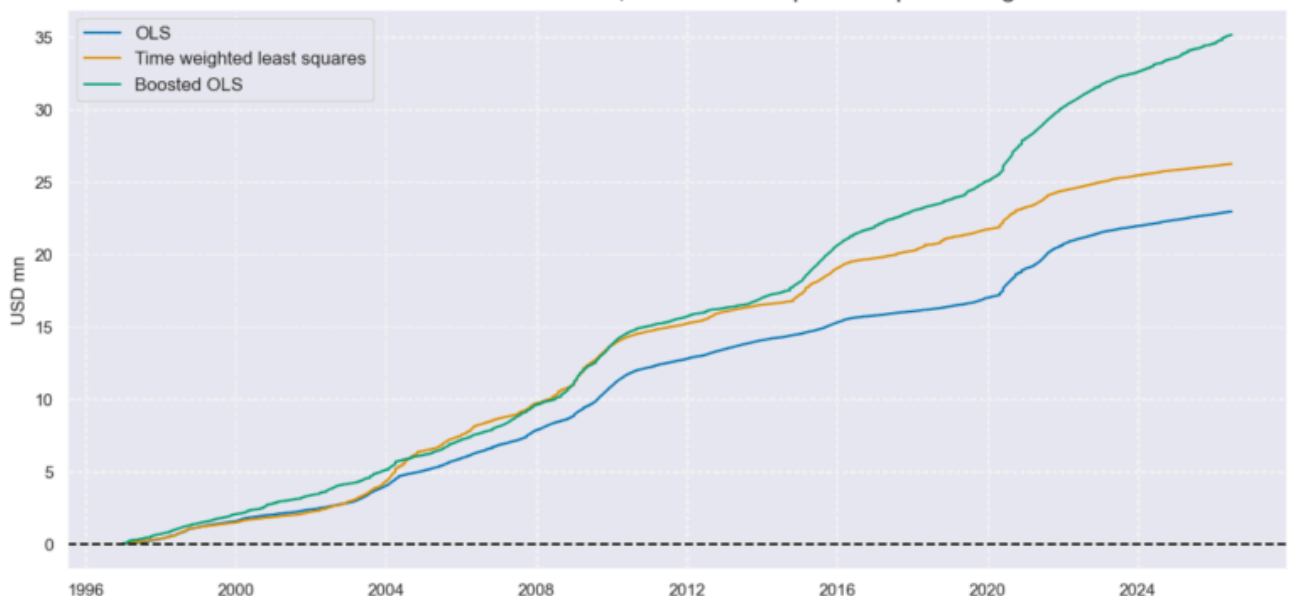
Simulated FX forward PnLs for 8 developed markets and USD250 million risk capital



A basic transaction cost analysis

For USD250 million in assets under management and an annualized PnL of between USD15 and 18 million, the trading and roll costs of the FX forward positions reduced economic gains only modestly, by 5-6% of the generated PnL. Also, the accumulation of these costs has been gradual and in accordance with signal strength. This partly reflects the simplistic assumption of stable transaction costs per ticket size, disregarding fluctuations in market conditions.

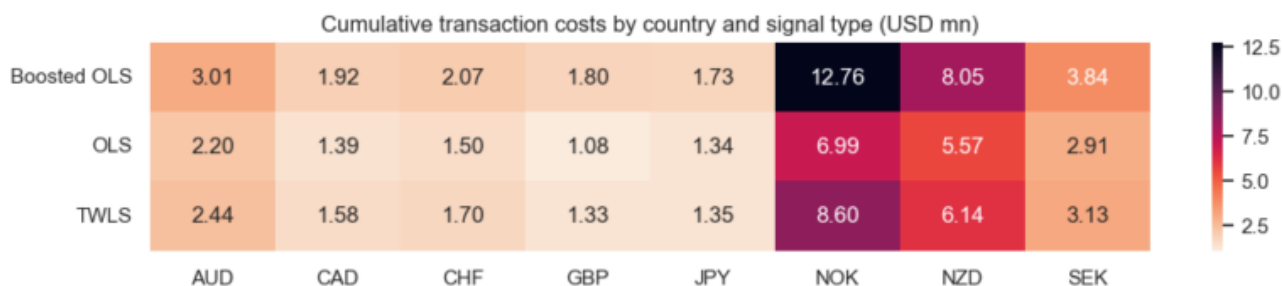
Cumulative transaction costs, for USD50 mn positions per unit signal



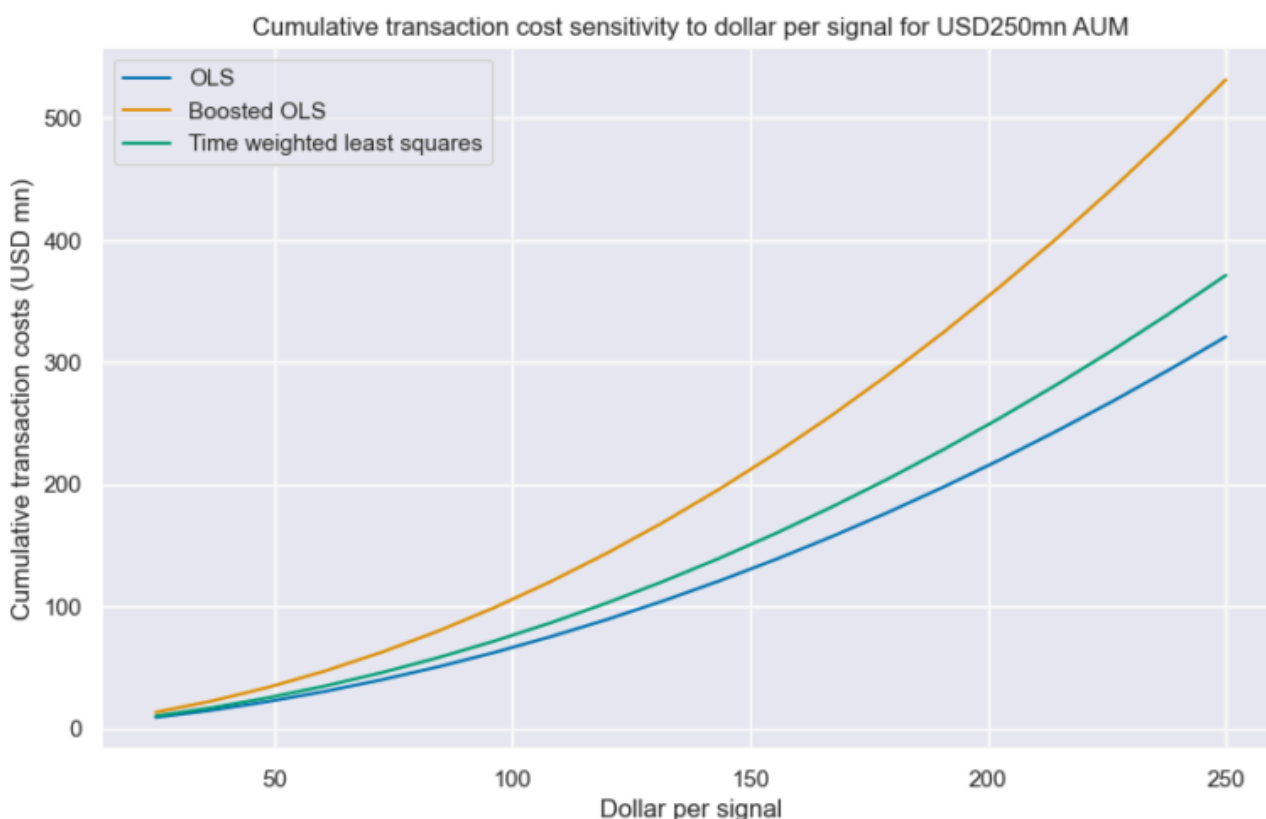
Time-variant trading costs and volatility targeting on the part of the strategy would change that picture, but also make analyses very sensitive to assumptions.

However, even with the simplistic assumption of constant size-dependent trading and roll costs, we can conduct important analyses to inform the size-risk capital that can be allocated to the strategy.

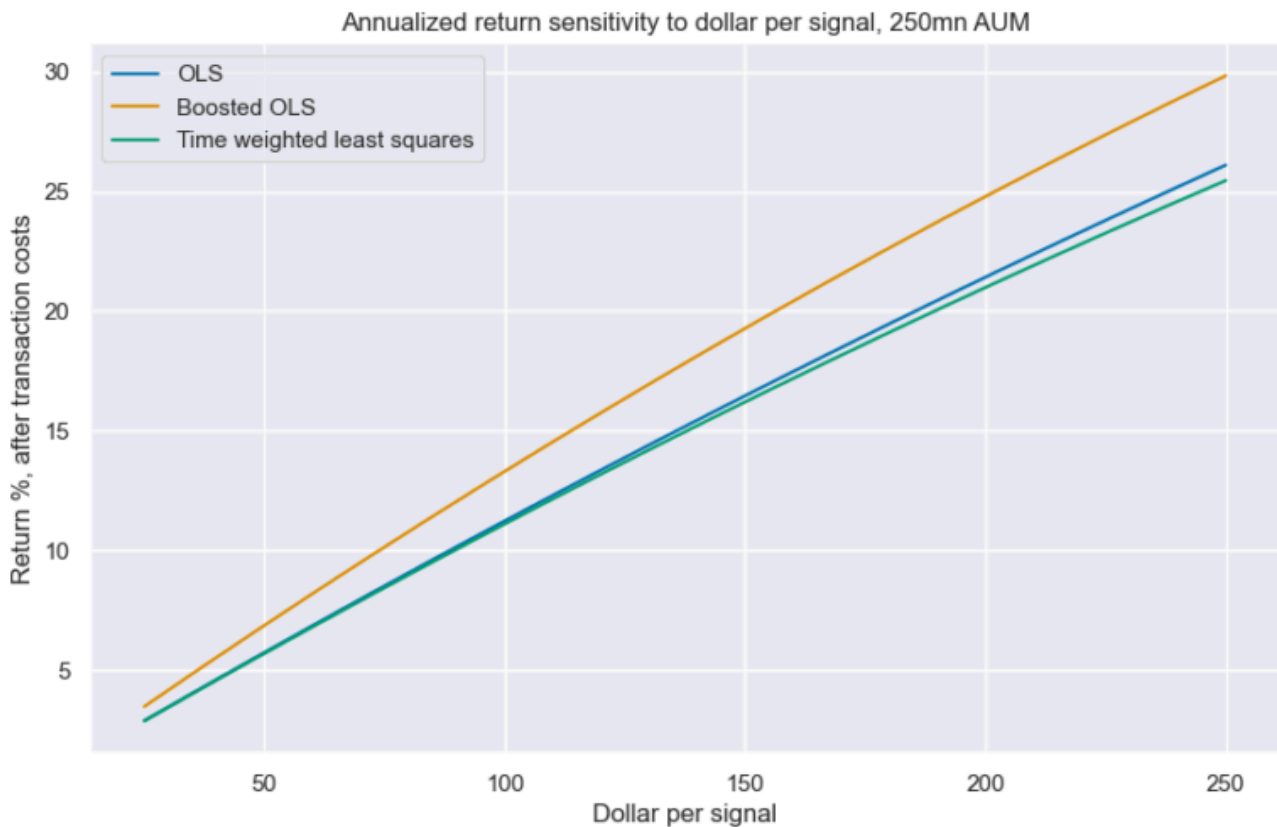
One important finding is that transaction costs are concentrated on two smaller markets: EURNOK and NZDUSD forwards. Together, they accounted for more than 50% of the total costs of the standard least-squares-based strategy, for example. This can be viewed as the price of diversification. For strategies with high assets under management, it may be more profitable to exclude the smaller markets from actual trading, while retaining their use in statistical learning and signal generation.



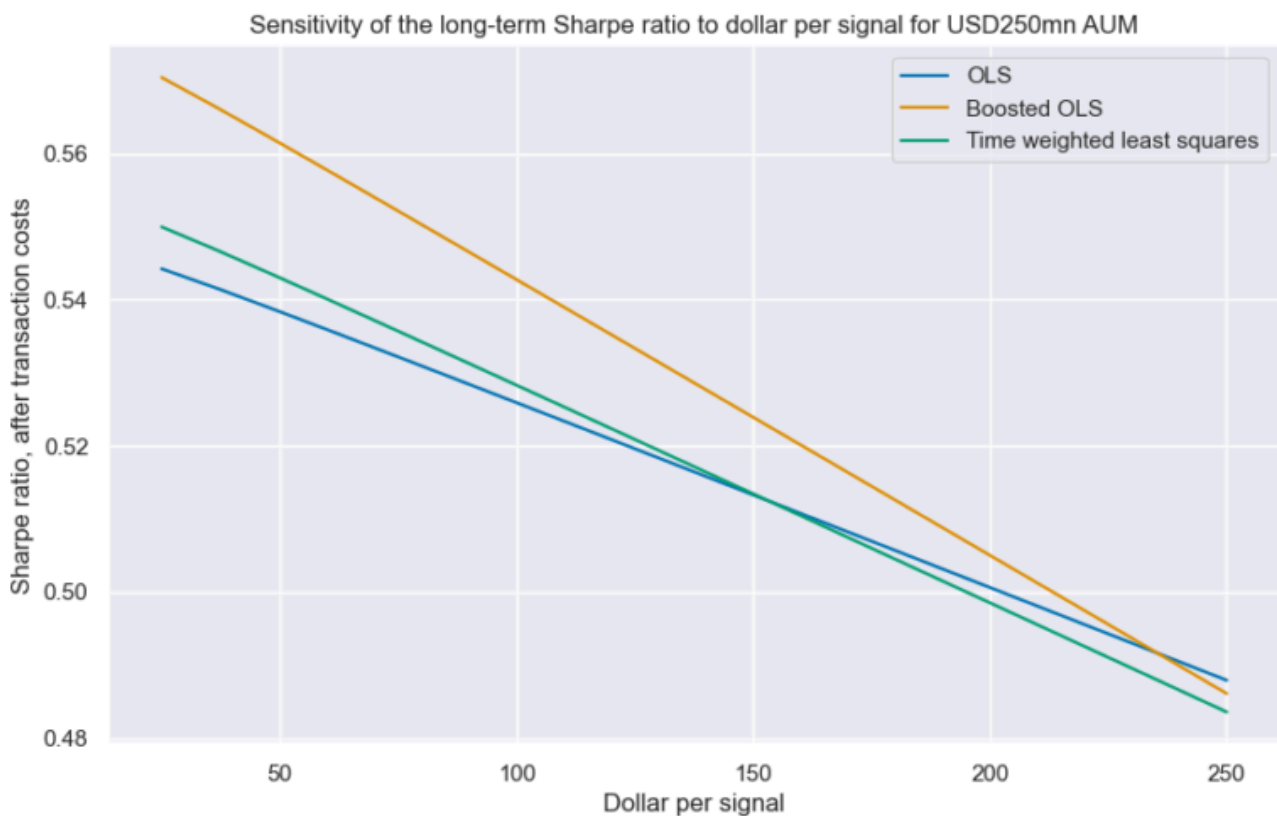
Transaction cost analysis, even in a simple basic form, is useful for estimating the effects of scale and leverage. The graphic below shows the impact of increasing leverage on cumulative transaction cost, highlighting that (i) costs are growing more than proportionally to scale and (ii) the increase in USD cost is greater for the “bolder” boosted regression-based signals.



The graphic below addresses the question of whether returns per AUM will keep increasing if we raise the strategy's leverage through higher USD position per unit signals. Importantly, the analysis suggests that the increase in return is less than linear but remains positive up to a ratio of USD 250 million per unit signal. This suggests that nominal PnLs could be raised to USD 75 million under the current strategy. This is a very important result that highlights an important feature of low-frequency macro strategies in liquid developed markets: their Sharpe and Sortino ratios may not be very high, but their nominal uncorrelated value generation can be very large.



The price of higher leverage is lower risk-adjusted percent returns. Our simple transaction cost analysis suggests that Sharpe ratios after transaction costs would decline from above 0.54-0.56 for USD50 million positions per unit signal to near 0.48 for USD250 million unit positions. This makes such strategies suitable for long-term investors who are willing to trade tight risk limits for a transparent systematic approach and high absolute returns.



Annex 1: Transaction cost prompt

The prompt below has been submitted to GPT-5.5 (thinking mode):

You are an FX transaction-cost analyst.

Estimate the distribution of realistic institutional bank-to-client trade sizes and associated transaction costs for 1-month FX forward trades in developed-market currencies.

For each currency, estimate both:

1. One-way 1M FX forward execution cost versus mid
2. Roll transaction cost excluding interest-rate carry

The goal is to estimate, for each currency:

- The median institutional bank-to-client 1M FX forward trade size
- The 90th percentile institutional bank-to-client 1M FX forward trade size
- The estimated one-way execution cost versus mid at the median trade size
- The estimated roll transaction cost excluding carry at the median trade size
- The estimated one-way execution cost versus mid at the 90th percentile trade size
- The estimated roll transaction cost excluding carry at the 90th percentile trade size

Express all trade sizes in USD-equivalent notional.

Express all costs in basis points of traded notional.

Use the following definitions:

- **One-way 1M FX forward execution cost versus mid** means the estimated cost for an institutional client to execute a new 1-month FX forward at a realistic bank-to-client price compared with the contemporaneous mid-market forward rate. This should include the dealer's effective bid-offer spread, client markup, liquidity premium, market-impact allowance, and normal execution slippage where relevant. This is a one-way execution cost versus mid, not the full bid-offer spread.
- **Roll transaction cost excluding carry** means the incremental transaction cost of rolling an existing 1M forward exposure into a new 1M forward position, excluding the carry implied by interest-rate differentials. It should capture the practical transaction cost of rolling, including forward-point bid-offer costs, market liquidity, dealer balance-sheet usage, funding/liquidity premia, and any additional spread typically charged to institutional clients.

Do not model the roll as two independent outright forward trades. In particular, do not assume a full close-out of the old forward versus mid plus a separate fresh outright forward execution versus mid. The intended concept is the practical transaction cost of extending the maturity of an existing exposure, not the cost of fully exiting and re-entering the

exposure.

Because the roll is primarily a maturity-extension transaction rather than a new outright risk-transfer trade, the estimated roll transaction cost excluding carry should normally be lower than the corresponding one-way outright 1M forward execution cost at the same trade size.

Estimate results for the following currencies: NOK, SEK, CAD, JPY, AUD, GBP, CHF, and NZD.

Use the following trading conventions:

- NOK, SEK, and CHF are traded versus EUR.
- CAD, JPY, AUD, and NZD are traded versus USD.
- GBP is traded versus an equally weighted basket of EUR and USD.
- All estimates should refer to 1-month FX forwards.
- Trade sizes should be expressed in USD-equivalent terms.

For each currency, estimate the median and 90th percentile trade sizes based on typical institutional activity, relative liquidity, turnover, market depth, dealer balance-sheet capacity, and the usual size distribution of 1-month FX forward trades.

Then estimate the one-way execution cost versus mid and the roll transaction cost excluding carry that would apply at each of those trade sizes.

Return the results in a table with the following columns:

- Currency
- Trading pair or basket
- Estimated median 1M forward trade size, USD-equivalent
- One-way execution cost versus mid at median trade size, bps
- Roll transaction cost excluding carry at median trade size, bps
- Estimated 90th percentile 1M forward trade size, USD-equivalent
- One-way execution cost versus mid at 90th percentile trade size, bps
- Roll transaction cost excluding carry at 90th percentile trade size, bps

After the table, briefly explain:

1. The assumptions used to estimate the 1M forward trade-size distribution.
2. The assumptions used to estimate one-way execution costs versus mid.

3. The assumptions used to estimate roll transaction costs, clearly separating roll transaction costs from interest-rate carry.
4. Which currencies are likely to have the lowest total transaction costs and why.
5. Which currencies are likely to have the highest total transaction costs and why.
6. How costs scale from the median trade size to the 90th percentile trade size.

Use rounded estimates rather than false precision. If uncertainty is material, provide ranges or clearly state that the estimates are approximate. Prioritize internal consistency across currencies, clear economic reasoning, and realistic institutional bank-to-client pricing rather than interbank screen spreads.

Annex 2: Macro-quantamental factors

These are the 10 macro-quantamental factors considered for the developed market FX forward trading with documentation links:

- **Relative GDP growth trends:** This is the difference between current nowcast GDP growth in the local currency area and the benchmark currency area. Higher relative growth indicates better economic competitiveness and warrants tighter monetary policy. The factor is represented by the JPMaQS category group "**Intuitive GDP growth estimates**," real-time estimates of recent GDP growth trends (% over a year ago, 3-month average) based on regressions using the latest available national accounts data and monthly-frequency activity data. It mimics common methods of market economists.
- **Relative private consumption growth:** This measures the strength in real household spending in the base economy versus the benchmark currency area. Stronger real spending argues for tighter monetary policy. The factor is based on real private consumption, percentage over a year ago, with a 3-month moving average filter (see "**Private consumption growth**").
- **Manufacturing confidence improvement:** These are changes in normalized headline measures of local-currency area manufacturing business sentiment. Generally, improving confidence in tradable goods industries signals stronger competitiveness and leads a positive news flow. The basis of the indicator is the JPMaQS category group "**Manufacturing confidence scores**". These are standardized and seasonally adjusted measures of manufacturing business confidence and their changes based on one or more surveys per country and currency area. For the analysis below, we consider averages of annualized 3-month-over-3-month and 6-month-over-6-month changes in survey scores.
- **Relative unemployment trends:** These are recent changes in seasonally adjusted unemployment rates in the local economy versus the benchmark currency area. Relative tightening of the labor market in the local currency area bodes for relative tightening of monetary policy. The basis of the indicator is the JPMaQS category group "**Labor market dynamics**," which contains real-time measures of changes in employment and unemployment. Here, we use average annualized changes of 3 months over 3 months, 6 months over 6 months, and over a year ago in seasonally adjusted unemployment rates.
- **Relative excess core CPI inflation:** This factor measures the excess of local core inflation versus the excess of core inflation in the benchmark currency area. Excess here means relative to the central bank's inflation target. If local inflation exceeds its target more than in the benchmark currency area, it is more likely that the local central bank will run a tighter monetary policy and tolerate currency strength. The factor is calculated by the JPMaQS

categories "**Consistent core CPI trends**" and "**Inflation targets**". Specifically, we use a weighted average 3-month-over-3-month, 6-month-over-6-month, and 12-month-over-12-month annualized changes in core CPI and subtract the currency area's effective inflation target from these.

- **Relative excess producer price inflation:** This is the difference between the excess producer price trend in the local economy and the benchmark currency area. Higher local currency area excess producer price growth indicates both pressures for tighter monetary policy and improving competitiveness. These price trends are part of the JPMaQS category group "**Producer price inflation**." Here, we use the standard (mostly manufacturing) producer price index and an economy-wide estimated output price index, which effectively is nowcast GDP deflator growth. For both, we observe excess inflation as the percentage change over a year ago, with a 3-month moving average filter.
- **Relative excess inflation expectations:** This is the difference between excess estimated inflation expectations in the local economy and the benchmark currency area. Higher relative inflation expectations across similar economies typically translate into tighter monetary policy and greater tolerance for currency strength. Here, inflation expectations refer to formulaic estimates of the JPMaQS category group "**Inflation expectations (Macrosynergy method)**." Here, we use an average of 1-year, 2-year, and 5-year inflation expectations.
- **Relative real interest rates:** This is the difference between various real interest rate in the local market and in the benchmark currency area. Higher real interest rates are indicative of higher implied central bank subsidies (or lower penalties) and higher risk premia. The real short-term rates are taken from the JPMaQS category "**Real interest rates**." In particular, we use the main 1-month money market rate and the 2-year and 5-year swap yields adjusted for formulaic inflation expectations.
- **International investment position trends:** Conceptually, these are changes in recorded external financial relations between residents and non-residents over the medium term. A rapid increase in net liabilities indicates large past capital inflows that may be hard to sustain and, hence, increases the risk of subsequent currency weakness. The basis of the indicator calculation is net international investment positions and (the negative of) international liabilities as % of GDP from the JPMaQS category group "**International investment position**." Here we look at trends in information states of the two ratios versus the average of the past two years and five years in percentage points of GDP, and take an average.
- **Terms-of-trade dynamics:** These are changes in export prices relative to changes in import prices. Improving terms of trade often precede economic outperformance, capital inflows, and positive economic news, all of which support currency strength. The basis of terms-of-trade dynamics here is the commodity-based terms-of-trade dynamics of the JPMaQS group "**Terms-of-trade**." Commodity-based terms of trade use commodity trade and price data alone and can be consistently updated in real-time at a daily frequency. The factor is an average of annualized changes of the latest month against a year ago, the latest month over the previous 12 months, and the latest week over the previous 4 weeks.

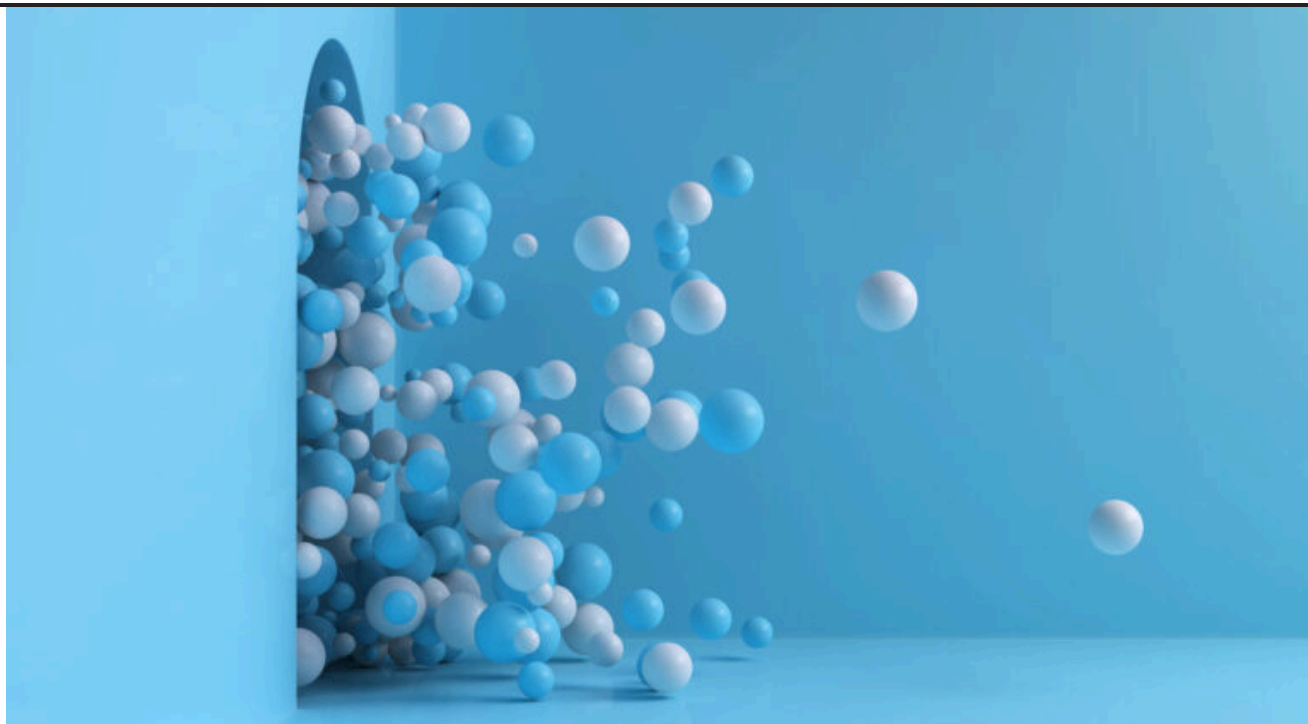
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